Integration Theory MATM39: Summary of Important Results

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Chapter 1

Sigma Algebras

Definition 1. *Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a* σ-algebra *on X if:*

(a) $X \in \mathcal{A}$,

(b) if $A \in \mathcal{A}$, *then* $A^c \in \mathcal{A}$,

(c) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, *then* $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$,

(d) if $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, *then* $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i, and by (c), $\bigcup_{i=1}^{\infty} A_i^c \in \mathcal{A}$. By (b) again, $\left(\bigcup_{i=1}^{\infty} A_i^c\right)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i^c\right)^c \in \mathcal{A}$$

$\square$

Proposition 1. *Let X be a set. Then the intersection of an arbitrary nonempty collection of σ-algebras on X is a σ-algebra on X.*

Proof. Let $\mathcal{C}$ be a nonempty collection of σ-algebras on X, and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X, we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ-algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^{\infty} \subset \mathcal{A}$, then $\{A_i\}_{i=1}^{\infty} \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ-algebra on X. $\square$

Corollary 1. *Let X be a set, and let $\mathcal{F}$ be a family of subsets of X. Then there exists a smallest σ-algebra on X that contains $\mathcal{F}$.*

Proof. Let $\mathcal{C}$ be the collection of all σ-algebras on X that contain $\mathcal{F}$. This collection is nonempty, since $\mathcal{P}(X)$ is a σ-algebra on X and contains $\mathcal{F}$.

Define

$$\sigma(\mathcal{F}) := \bigcap_{\mathcal{A}\in\mathcal{C}} \mathcal{A}.$$

By the previous proposition, $\sigma(\mathcal{F})$ is a σ-algebra on X. It contains $\mathcal{F}$, since every $\mathcal{A} \in \mathcal{C}$ contains $\mathcal{F}$. Moreover, if $\mathcal{B}$ is any σ-algebra on X that contains $\mathcal{F}$, then $\mathcal{B} \in \mathcal{C}$, and therefore

$$\sigma(\mathcal{F}) \subset \mathcal{B}.$$

Thus $\sigma(\mathcal{F})$ is the smallest σ-algebra on X containing $\mathcal{F}$. □

Definition 2. *The* Borel σ-algebra *on $\mathbb{R}^d$ is the σ-algebra generated by the open subsets of $\mathbb{R}^d$. It is denoted by $\mathcal{B}(\mathbb{R}^d)$. Its elements are called* Borel sets.

Proposition 2. *The Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is generated by each of the following collections:*

(a) the collection of all closed subsets of $\mathbb{R}$,

(b) the collection of all intervals of the form $(-\infty, b]$,

(c) the collection of all intervals of the form $(a, b]$.

Proof. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ denote the σ-algebras generated by the collections in (a), (b), and (c), respectively.

Since $\mathcal{B}(\mathbb{R})$ contains all open sets and is closed under complements, it contains all closed sets. Hence

$$\mathcal{B}(\mathbb{R}) \supset \mathcal{B}_1.$$

Every set $(-\infty, b]$ is closed, so $\mathcal{B}_1 \supset \mathcal{B}_2$. Moreover,

$$(a, b] = (-\infty, b] \cap (-\infty, a]^c,$$

so $\mathcal{B}_2 \supset \mathcal{B}_3$.

Finally, let $G \subset \mathbb{R}$ be open. For each $x \in G$, since G is open, there exist rational numbers $a, b \in \mathbb{Q}$ such that

$$x \in (a, b] \subset G.$$

Hence

$$G = \bigcup_{\substack{a,b\in\mathbb{Q} \\ (a,b]\subset G}} (a, b].$$

This is a countable union, since $\mathbb{Q}^2$ is countable. Therefore $G \in \mathcal{B}_3$. Thus every open set belongs to $\mathcal{B}_3$, so

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{B}_3.$$

Therefore all four σ-algebras are equal. □

Measures

Definition 3. *Let X be a set and $\mathcal{A}$ a σ-algebra on X. A function $\mu : \mathcal{A} \to [0, \infty]$ is called a* measure *if*

(a) $\mu(\emptyset) = 0$,

(b) for every $\{A_i\}_{i=1}^\infty$ *of pairwise disjoint sets in* $\mathcal{A}$, $\mu(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \mu(A_i)$.

Proposition 3 (Proposition 1.2.5)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space.*

(a) If $\{A_k\}$ is an increasing sequence in $\mathcal{A}$, then

$$\mu\left(\bigcup_{k=1}^\infty A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

(b) If $\{A_k\}$ is a decreasing sequence in $\mathcal{A}$ and $\mu(A_n) < \infty$ for some n, then

$$\mu\left(\bigcap_{k=1}^\infty A_k\right) = \lim_{k\to\infty} \mu(A_k).$$

Proof. (a) Define $B_1 = A_1$ and $B_k = A_k \setminus A_{k-1}$ for $k \geq 2$. Then $\{B_k\}$ are disjoint and $\bigcup_k A_k = \bigcup_k B_k$. Hence

$$\mu\left(\bigcup_k A_k\right) = \sum_k \mu(B_k) = \lim_{k\to\infty} \sum_{i=1}^k \mu(B_i) = \lim_{k\to\infty} \mu(A_k).$$

(b) Since $\mu(A_n) < \infty$ for some n, we may replace $\{A_k\}$ by the tail $\{A_{k+n-1}\}$ and assume $\mu(A_1) < \infty$. Let $C_k = A_1 \setminus A_k$. Then $\{C_k\}$ is increasing and

$$\bigcup_k C_k = A_1 \setminus \bigcap_k A_k.$$

By (a),

$$\mu(A_1 \setminus \bigcap_k A_k) = \lim_{k\to\infty} \mu(C_k) = \lim_{k\to\infty} \big(\mu(A_1) - \mu(A_k)\big),$$

which gives $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$. □

Outer Measures

Definition 4. *Let X be a set. An* outer measure *on X is a function $\mu^* : \mathcal{P}(X) \to [0, \infty]$ such that*

(a) $\mu^*(\emptyset) = 0$,

(b) if $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$,

(c) for any sequence $\{A_n\}$ of subsets of X, $\mu^*(\bigcup_{n=1}^\infty A_n) \leq \sum_{n=1}^\infty \mu^*(A_n)$.

Definition 5. *The* Lebesgue outer measure *on* $\mathbb{R}$ *is the function* $\lambda^* : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ *defined by*

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : \ A \subset \bigcup_{i=1}^{\infty} (a_i, b_i) \right\}.$$

Definition 6. *The* Lebesgue outer measure *on* $\mathbb{R}^d$ *is defined by*

$$\lambda^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \mathrm{vol}(R_i) : \ A \subset \bigcup_{i=1}^{\infty} R_i \right\},$$

where each R_i *is a bounded open rectangle in* $\mathbb{R}^d$ *and*

$$\mathrm{vol}(I_1 \times \cdots \times I_d) = \prod_{j=1}^{d} |I_j|.$$

Definition 7. *Let* μ^* *be an outer measure on* X*. A set* $B \subset X$ *is called* μ^*-measurable *if for all* $A \subset X$*,*

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^c).$$

Remark 1. *Outer measures are defined on all subsets of* X*, but are only subadditive in general. Carathéodory's criterion selects a class of sets on which* μ^* *becomes additive. This allows us to restrict* μ^* *to a* σ*-algebra and obtain a genuine measure (in particular, the Lebesgue measure).*

Theorem 1. *Let* X *be a set and* μ^* *an outer measure on* X*. Let*

$$\mathcal{M}_{\mu^*} := \{B \subset X : B \text{ is } \mu^*\text{-measurable}\}.$$

Then:

(a) $\mathcal{M}_{\mu^*}$ *is a* σ*-algebra,*

(b) the restriction of μ^* *to* $\mathcal{M}_{\mu^*}$ *is a measure.*

Proof sketch. One shows that $\mathcal{M}_{\mu^*}$ is closed under complements directly from the definition. Closure under finite unions follows by applying the defining identity to B_1 and B_2 and decomposing sets appropriately.

Closure under countable unions is obtained by first proving the result for finite disjoint unions and then passing to the limit using subadditivity of μ^*.

Finally, countable additivity of μ^* on $\mathcal{M}_{\mu^*}$ follows by applying the defining property to disjoint measurable sets and using the previous decomposition. □

Remark 2. *Applying this theorem to the Lebesgue outer measure* λ^* *on* $\mathbb{R}^d$ *produces a* σ*-algebra* $\mathcal{M}_{\lambda^*}$*, called the* Lebesgue σ-algebra*. The restriction of* λ^* *to this* σ*-algebra is the Lebesgue measure. Thus, outer measure together with Carathéodory's criterion provides a systematic way to construct Lebesgue measure.*

Example 1 (Cantor set)**.** *Let* $K_0 = [0, 1]$*. Construct* K_n *by removing the open middle third of each interval in* K_{n-1}*. The* Cantor set *is* $K = \bigcap_{n=0}^{\infty} K_n$*.*

Proposition 4. *The Cantor set* K *is compact, uncountable, and has Lebesgue measure zero.*

Idea. Each K_n is a finite union of closed intervals, hence compact, and $K = \bigcap_n K_n$ is compact. One shows K has the cardinality of the continuum via ternary expansions.

Moreover, $\lambda(K_n) = (2/3)^n$, so

$$\lambda(K) \leq \lambda(K_n) = (2/3)^n \to 0,$$

hence $\lambda(K) = 0$. □

Theorem 2. *There exists a subset of* $(0, 1)$ *that is not Lebesgue measurable.*

Idea. Define $x \sim y$ if $x - y \in \mathbb{Q}$ and choose one representative from each equivalence class (using the axiom of choice) to form a set $E \subset (0, 1)$.

Consider the translates $E + r_n$ by rational numbers r_n. These sets are disjoint and their union covers $(0, 1)$ up to inclusion. Translation invariance and countable additivity would force a contradiction, so E cannot be measurable. □

Remark 3. *Not every subset of* $\mathbb{R}$ *can be assigned a consistent measure: the existence of non-measurable sets shows that countable additivity and translation invariance cannot hold on all subsets.*

Therefore, one must restrict attention to a σ*-algebra (such as the Lebesgue* σ*-algebra) on which a well-defined measure exists.*

Chapter 2

Definition 8. *Let* $(X, \mathcal{A}, \mu)$ *be a measure space. A nonnegative simple measurable function is a function of the form*

$$f = \sum_{i=1}^{n} a_i \mathbf{1}_{A_i},$$

where $a_i \geq 0$ *and* $A_i \in \mathcal{A}$ *are disjoint. Its integral is defined by*

$$\int f \, d\mu := \sum_{i=1}^{n} a_i \mu(A_i).$$

Definition 9. *Let* $f : X \to [0, \infty]$ *be measurable. The integral of* f *is defined by*

$$\int f \, d\mu := \sup \left\{ \int g \, d\mu : g \in \mathcal{S}_+, \ g \leq f \right\},$$

where $\mathcal{S}_+$ *denotes the set of nonnegative simple measurable functions.*

Proposition 5. *Every nonnegative measurable function* $f : X \to [0, \infty]$ *is the pointwise limit of an increasing sequence of nonnegative simple measurable functions.*

Reference. This is Proposition 2.1.8 in Cohn. It guarantees that nonnegative measurable functions can be approximated from below by simple functions. □

Proposition 6. *If* $f_n \in \mathcal{S}_+$*,* $f_n \uparrow f$*, and* f *is nonnegative measurable, then*

$$\int f \, d\mu = \lim_{n \to \infty} \int f_n \, d\mu.$$

Reference. This is Proposition 2.3.3 in Cohn. It shows that the supremum definition agrees with taking limits along increasing simple approximations. Proposition 2.3.2 is the corresponding result when the limiting function is itself simple. □

Remark 4. *Thus the integral is first defined directly for nonnegative simple functions. For arbitrary nonnegative measurable functions, it is defined as the supremum of the integrals of all simple functions below it. Propositions 2.1.8 and 2.3.3 show that this construction is compatible with approximation from below.*

Definition 10. *Let $f : X \to [-\infty, \infty]$ be measurable. Define*

$$f^+ := \max(f, 0), \qquad f^- := \max(-f, 0).$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

If at least one of $\int f^+ \, d\mu$ and $\int f^- \, d\mu$ is finite, then $\int f \, d\mu$ is said to exist, *and is defined by*

$$\int f \, d\mu := \int f^+ \, d\mu - \int f^- \, d\mu.$$

If both $\int f^+ \, d\mu$ and $\int f^- \, d\mu$ are finite, then f is called integrable.

Definition 11. *If $A \in \mathcal{A}$, then f is called* integrable over A *if $f\mathbf{1}_A$ is integrable. In that case,*

$$\int_A f \, d\mu := \int f\mathbf{1}_A \, d\mu.$$

Proposition 7 (Proposition 2.3.2)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f \in \mathcal{S}_+$, and let $\{f_n\}$ be an increasing sequence in $\mathcal{S}_+$ such that*

$$f = \lim_{n\to\infty} f_n$$

pointwise on X. Then

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Proof. By monotonicity,

$$\int f_1 \, d\mu \le \int f_2 \, d\mu \le \cdots \le \int f \, d\mu,$$

so

$$\lim_{n\to\infty} \int f_n \, d\mu \le \int f \, d\mu.$$

We prove the reverse inequality. Let $0 < \varepsilon < 1$. Write

$$f = \sum_{i=1}^{k} a_i \mathbf{1}_{A_i},$$

where $a_i > 0$ and the sets $A_i \in \mathcal{A}$ are disjoint. For each n and i, define

$$A(n, i) := \{x \in A_i : f_n(x) \ge (1 - \varepsilon)a_i\}.$$

Then $A(n,i) \in \mathcal{A}$, the sequence $\{A(n,i)\}_{n=1}^{\infty}$ is increasing, and

$$A_i = \bigcup_{n=1}^{\infty} A(n,i).$$

Define

$$g_n := \sum_{i=1}^{k} (1-\varepsilon) a_i \mathbf{1}_{A(n,i)}.$$

Then $g_n \in \mathcal{S}_+$ and $g_n \le f_n$. Hence

$$\int g_n \, d\mu \le \int f_n \, d\mu.$$

By Proposition 1.2.5,

$$\lim_{n\to\infty} \int g_n \, d\mu = \lim_{n\to\infty} \sum_{i=1}^{k} (1-\varepsilon) a_i \mu(A(n,i)) = \sum_{i=1}^{k} (1-\varepsilon) a_i \mu(A_i) = (1-\varepsilon) \int f \, d\mu.$$

Therefore

$$(1-\varepsilon) \int f \, d\mu \le \lim_{n\to\infty} \int f_n \, d\mu.$$

Since $\varepsilon > 0$ was arbitrary,

$$\int f \, d\mu \le \lim_{n\to\infty} \int f_n \, d\mu.$$

Thus

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

□

Proposition 8 (Proposition 2.3.3). *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $f : X \to [0, \infty]$ be measurable, and let $\{f_n\}$ be an increasing sequence of functions in $\mathcal{S}_+$ such that*

$$f = \lim_{n\to\infty} f_n$$

pointwise on X. Then

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Proof. By monotonicity of the integral,

$$\int f_1 \, d\mu \le \int f_2 \, d\mu \le \cdots \le \int f \, d\mu.$$

Hence

$$\lim_{n\to\infty} \int f_n \, d\mu \le \int f \, d\mu.$$

For the reverse inequality, let $g \in \mathcal{S}_+$ with $g \le f$. Then $\{g \wedge f_n\}$ is an increasing sequence in $\mathcal{S}_+$ and

$$g = \lim_{n\to\infty} (g \wedge f_n).$$

By Proposition 2.3.2,
$$\int g\,d\mu = \lim_{n\to\infty} \int (g \wedge f_n)\,d\mu.$$
Since $g \wedge f_n \le f_n$, we arrive at
$$\int f\,d\mu = \sup\left\{ \int g\,d\mu : g \in \mathcal{S}_+,\ g \le f \right\} \le \lim_{n\to\infty} \int f_n\,d\mu.$$
□

Proposition 9 (Proposition 2.3.9)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, g : X \to [-\infty, \infty]$ be measurable functions such that $f = g$ almost everywhere. If either $\int f\,d\mu$ or $\int g\,d\mu$ exists, then both exist and*
$$\int f\,d\mu = \int g\,d\mu.$$

Proof. First suppose $f, g \ge 0$. Let
$$A = \{x \in X : f(x) \ne g(x)\}.$$
Then $\mu(A) = 0$. Define
$$h(x) = \begin{cases} +\infty, & x \in A, \\ 0, & x \notin A. \end{cases}$$
Since $\mu(A) = 0$, define
$$h_n := n\mathbf{1}_A.$$
Then $h_n \in \mathcal{S}_+$, $h_n \uparrow h$, and
$$\int h_n\,d\mu = n\mu(A) = 0.$$
By Proposition 2.3.3,
$$\int h\,d\mu = \lim_{n\to\infty} \int h_n\,d\mu = 0.$$
Since $f \le g + h$ and $g \le f + h$, monotonicity gives
$$\int f\,d\mu \le \int g\,d\mu + \int h\,d\mu = \int g\,d\mu$$
and similarly
$$\int g\,d\mu \le \int f\,d\mu.$$
Thus $\int f\,d\mu = \int g\,d\mu$.

For general f, g, apply the nonnegative case to f^+, g^+ and to f^-, g^-. □

Proposition 10 (Proposition 2.3.10)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f : X \to [0, \infty]$ be measurable. If $t > 0$ and*
$$A_t := \{x \in X : f(x) \ge t\},$$
then
$$\mu(A_t) \le \frac{1}{t} \int_{A_t} f\,d\mu \le \frac{1}{t} \int f\,d\mu.$$

Proof. On A_t we have $f \geq t$, hence

$$0 \leq t\mathbf{1}_{A_t} \leq f\mathbf{1}_{A_t} \leq f.$$

By monotonicity of the integral,

$$\int t\mathbf{1}_{A_t}\, d\mu \leq \int_{A_t} f\, d\mu \leq \int f\, d\mu.$$

Since

$$\int t\mathbf{1}_{A_t}\, d\mu = t\mu(A_t),$$

division by t gives

$$\mu(A_t) \leq \frac{1}{t}\int_{A_t} f\, d\mu \leq \frac{1}{t}\int f\, d\mu.$$

□

Theorem 3 (Monotone Convergence Theorem)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \ldots$ be $[0, \infty]$-valued measurable functions on X. Suppose that*

$$f_1 \leq f_2 \leq \cdots$$

and

$$f = \lim_{n\to\infty} f_n$$

hold almost everywhere. Then

$$\int f\, d\mu = \lim_{n\to\infty} \int f_n\, d\mu.$$

Proof. First assume that $f_n \uparrow f$ pointwise everywhere. By monotonicity of the integral,

$$\int f_1\, d\mu \leq \int f_2\, d\mu \leq \cdots \leq \int f\, d\mu.$$

Hence

$$\lim_{n\to\infty} \int f_n\, d\mu \leq \int f\, d\mu.$$

We prove the reverse inequality. For each n, choose an increasing sequence $\{g_{n,k}\}_{k=1}^{\infty}$ of nonnegative simple functions such that

$$g_{n,k} \uparrow f_n.$$

This is possible by Proposition 2.1.8. Define

$$h_n := \max(g_{1,n}, g_{2,n}, \ldots, g_{n,n}).$$

Then $\{h_n\}$ is an increasing sequence of nonnegative simple functions, $h_n \leq f_n$, and

$$h_n \uparrow f.$$

Therefore, by Proposition 2.3.3,

$$\int f\, d\mu = \lim_{n\to\infty} \int h_n\, d\mu.$$

Since $h_n \le f_n$, monotonicity gives

$$\int h_n \, d\mu \le \int f_n \, d\mu.$$

Thus

$$\int f \, d\mu = \lim_{n\to\infty} \int h_n \, d\mu \le \lim_{n\to\infty} \int f_n \, d\mu.$$

Combining the two inequalities gives

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

If the assumptions hold only almost everywhere, choose a null set N outside which they hold. Apply the first part to $f_n \mathbf{1}_{N^c}$ and $f\mathbf{1}_{N^c}$. Since changing a function on a null set does not change its integral, the result follows. □

Theorem 4 (Fatou's Lemma)**.** *Let* $(X, \mathcal{A}, \mu)$ *be a measure space, and let* $\{f_n\}$ *be a sequence of* $[0, \infty]$*-valued measurable functions on* X*. Then*

$$\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

Proof. For each n, define

$$g_n := \inf_{k\ge n} f_k.$$

By Proposition 2.1.5 each g_n is measurable and

$$g_1 \le g_2 \le \cdots .$$

Moreover,

$$\lim_{n\to\infty} g_n = \liminf_{n\to\infty} f_n.$$

By the Monotone Convergence Theorem,

$$\int \liminf_{n\to\infty} f_n \, d\mu = \int \lim_{n\to\infty} g_n \, d\mu = \lim_{n\to\infty} \int g_n \, d\mu.$$

Since $g_n \le f_k$ for every $k \ge n$, we have

$$\int g_n \, d\mu \le \inf_{k\ge n} \int f_k \, d\mu.$$

Therefore

$$\lim_{n\to\infty} \int g_n \, d\mu \le \lim_{n\to\infty} \inf_{k\ge n} \int f_k \, d\mu = \liminf_{n\to\infty} \int f_n \, d\mu.$$

Hence

$$\int \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int f_n \, d\mu.$$

□

Theorem 5 (Dominated Convergence Theorem)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space. Let $g : X \to [0, \infty]$ be integrable, and let $f, f_1, f_2, \ldots$ be $[-\infty, \infty]$-valued measurable functions on X such that*

$$f_n \to f$$

almost everywhere, and

$$|f_n| \le g$$

almost everywhere for every n. Then f and $f_1, f_2, \ldots$ are integrable, and

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Proof. First fix n. Since $|f_n| \le g$ almost everywhere, Proposition 2.3.9 says that changing functions on a null set does not change their integrals. Hence we may treat the inequality as holding everywhere for the purpose of integration.

Since $0 \le |f_n| \le g$, monotonicity of the integral, Proposition 2.3.4(c), gives

$$\int |f_n| \, d\mu \le \int g \, d\mu < \infty.$$

Thus $|f_n|$ is integrable. Proposition 2.3.8 says that a measurable function is integrable if and only if its absolute value is integrable. Therefore f_n is integrable.

It remains to prove that f is integrable. Since $f_n \to f$ almost everywhere and $|f_n| \le g$ almost everywhere for every n, we get

$$|f| \le g$$

almost everywhere. Again, by Proposition 2.3.9, we may ignore the null set where this inequality may fail. Since $0 \le |f| \le g$, monotonicity of the integral, Proposition 2.3.4(c), gives

$$\int |f| \, d\mu \le \int g \, d\mu < \infty.$$

Thus $|f|$ is integrable. By Proposition 2.3.8, since a function is integrable if and only if its absolute value is integrable, f is integrable.

We now prove that

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

First assume that

$$f_n(x) \to f(x), \qquad |f_n(x)| \le g(x), \qquad g(x) < \infty$$

hold for every $x \in X$.

Then $g + f_n \ge 0$ and $g + f \ge 0$. Since $g + f_n \to g + f$, Fatou's Lemma gives

$$\int (g + f) \, d\mu \le \liminf_{n\to\infty} \int (g + f_n) \, d\mu.$$

By linearity and since $\int g \, d\mu < \infty$,

$$\int g \, d\mu + \int f \, d\mu \le \int g \, d\mu + \liminf_{n\to\infty} \int f_n \, d\mu.$$

Hence

$$\int f \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

Similarly, $g - f_n \geq 0$ and $g - f \geq 0$. Since $g - f_n \to g - f$, Fatou's Lemma gives

$$\int (g - f) \, d\mu \leq \liminf_{n\to\infty} \int (g - f_n) \, d\mu.$$

Again using linearity,

$$\int g \, d\mu - \int f \, d\mu \leq \int g \, d\mu - \limsup_{n\to\infty} \int f_n \, d\mu.$$

Thus

$$\limsup_{n\to\infty} \int f_n \, d\mu \leq \int f \, d\mu.$$

Combining the two inequalities gives

$$\limsup_{n\to\infty} \int f_n \, d\mu \leq \int f \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu.$$

Therefore

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Now suppose the assumptions hold only almost everywhere. Let $N \in \mathcal{A}$ be a null set containing all points where one of the following fails:

$$f_n(x) \to f(x), \qquad |f_n(x)| \leq g(x) \text{ for some } n, \qquad g(x) < \infty.$$

The last condition may be included since g is integrable, so $g < \infty$ almost everywhere.

On N^c, all assumptions hold everywhere. Therefore the first part applies to the functions $f_n \mathbf{1}_{N^c}$, $f \mathbf{1}_{N^c}$, and $g \mathbf{1}_{N^c}$, giving

$$\int f \mathbf{1}_{N^c} \, d\mu = \lim_{n\to\infty} \int f_n \mathbf{1}_{N^c} \, d\mu.$$

Since $f \mathbf{1}_{N^c} = f$ almost everywhere and $f_n \mathbf{1}_{N^c} = f_n$ almost everywhere, Proposition 2.3.9 gives

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

$\square$

Definition 12. *Let $f : [a, b] \to \mathbb{R}$ be bounded, and let*

$$\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$$

be a partition of $[a, b]$. Define

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x), \qquad M_i = \sup_{x \in [x_{i-1}, x_i]} f(x).$$

The lower sum *and* upper sum *of f with respect to $\mathcal{P}$ are*

$$l(f, \mathcal{P}) = \sum_{i=1}^{n} m_i (x_i - x_{i-1}),$$

and

$$u(f, \mathcal{P}) = \sum_{i=1}^{n} M_i (x_i - x_{i-1}).$$

Definition 13. *The* lower Riemann integral *and* upper Riemann integral *of f over $[a,b]$ are defined by*

$$\underline{\int_a^b} f = \sup_{\mathcal{P}} l(f,\mathcal{P}), \qquad \overline{\int_a^b} f = \inf_{\mathcal{P}} u(f,\mathcal{P}).$$

If

$$\underline{\int_a^b} f = \overline{\int_a^b} f,$$

then f is called Riemann integrable *on $[a,b]$, and the common value is denoted by*

$$\int_a^b f(x)\,dx.$$

Lemma 1 (Lemma 2.5.1)**.** *A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann integrable if and only if for every $\varepsilon > 0$ there exists a partition $\mathcal{P}$ of $[a,b]$ such that*

$$u(f,\mathcal{P}) - l(f,\mathcal{P}) < \varepsilon.$$

Theorem 6. *Let $[a,b]$ be a closed bounded interval, and let $f : [a,b] \to \mathbb{R}$ be bounded. Then:*

(a) f is Riemann integrable if and only if f is continuous almost everywhere on $[a,b]$;

(b) if f is Riemann integrable, then f is Lebesgue integrable and

$$(R)\int_a^b f(x)\,dx = (L)\int_{[a,b]} f\,d\lambda.$$

Proof sketch. Assume first that f is Riemann integrable. Choose partitions $\mathcal{P}_n$ such that

$$u(f,\mathcal{P}_n) - l(f,\mathcal{P}_n) \to 0,$$

and refine them if necessary so that $\mathcal{P}_{n+1}$ refines $\mathcal{P}_n$.

Define simple functions g_n and h_n by letting g_n be the infimum of f on each interval of $\mathcal{P}_n$, and h_n the supremum of f on each such interval. Then

$$g_n \le f \le h_n,$$

with $\{g_n\}$ increasing and $\{h_n\}$ decreasing. Moreover,

$$\int_{[a,b]} g_n\,d\lambda = l(f,\mathcal{P}_n), \qquad \int_{[a,b]} h_n\,d\lambda = u(f,\mathcal{P}_n).$$

Let

$$g = \lim_{n\to\infty} g_n, \qquad h = \lim_{n\to\infty} h_n.$$

Since the lower and upper sums have the same limit, we get

$$\int_{[a,b]} (h-g)\,d\lambda = 0.$$

Since $h - g \ge 0$, this implies $h = g$ almost everywhere. Outside the partition points, equality $g(x) = h(x)$ forces f to be continuous at x. Hence f is continuous almost everywhere. Also, since $g \le f \le h$ and $g = h$ almost everywhere, f agrees almost everywhere with the measurable function g. Therefore f is Lebesgue measurable and integrable, and the Riemann and Lebesgue integrals agree.

Conversely, suppose that f is continuous almost everywhere. Take partitions $\mathcal{P}_n$ into equal subintervals and define g_n, h_n as above. Then

$$h_n - g_n \to 0$$

almost everywhere. Since f is bounded, the functions $h_n - g_n$ are dominated by an integrable constant. By the Dominated Convergence Theorem,

$$\int_{[a,b]} (h_n - g_n)\, d\lambda \to 0.$$

Thus

$$u(f, \mathcal{P}_n) - l(f, \mathcal{P}_n) \to 0.$$

By the Riemann criterion, f is Riemann integrable. □

Proposition 11 (Triangle inequality for integrals)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f : X \to [-\infty, \infty]$ be measurable. Then f is integrable if and only if $|f|$ is integrable. In that case,*

$$\left| \int f\, d\mu \right| \le \int |f|\, d\mu.$$

Proof. Recall that

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Thus f is integrable if and only if both f^+ and f^- are integrable, which is equivalent to $|f|$ being integrable. If f is integrable, then

$$\left| \int f\, d\mu \right| = \left| \int f^+\, d\mu - \int f^-\, d\mu \right| \le \int f^+\, d\mu + \int f^-\, d\mu = \int |f|\, d\mu.$$

□

Chapter 3

Proposition 12 (Proposition 3.1.3)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \ldots$ be real-valued measurable functions on X. If $\mu(X) < \infty$ and*

$$f_n \to f$$

almost everywhere, then $f_n \to f$ in measure.

Proof. We must show that for every $\varepsilon > 0$,

$$\mu\big(\{x \in X : |f_n(x) - f(x)| > \varepsilon\}\big) \to 0.$$

Fix $\varepsilon > 0$ and define

$$A_n := \{x \in X : |f_n(x) - f(x)| > \varepsilon\}, \qquad B_n := \bigcup_{k=n}^{\infty} A_k.$$

Then $\{B_n\}$ is a decreasing sequence of measurable sets. Moreover,

$$\bigcap_{n=1}^{\infty} B_n \subset \{x \in X : f_n(x) \text{ does not converge to } f(x)\}.$$

Since $f_n \to f$ almost everywhere, this intersection has measure zero.

Because $\mu(X) < \infty$ and $B_1 \subset X$, we have $\mu(B_1) < \infty$. Hence, by continuity from above of measures, Proposition 1.2.5,

$$\mu(B_n) \to \mu\left(\bigcap_{n=1}^{\infty} B_n\right) = 0.$$

Since $A_n \subset B_n$, we get

$$0 \leq \mu(A_n) \leq \mu(B_n) \to 0.$$

Therefore,

$$\mu\big(\{x \in X : |f_n(x) - f(x)| > \varepsilon\}\big) \to 0,$$

so $f_n \to f$ in measure. $\square$

Proposition 13. *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \ldots$ be real-valued measurable functions on X. If $f_n \to f$ in measure, then there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \to f$ almost everywhere.*

Proof. Since $f_n \to f$ in measure, for each k we can choose $n_k > n_{k-1}$ such that

$$\mu\left(\left\{x \in X : |f_{n_k}(x) - f(x)| > \frac{1}{k}\right\}\right) \leq \frac{1}{2^k}.$$

Define

$$A_k := \left\{x \in X : |f_{n_k}(x) - f(x)| > \frac{1}{k}\right\}.$$

Then

$$\sum_{k=1}^{\infty} \mu(A_k) \leq \sum_{k=1}^{\infty} \frac{1}{2^k} < \infty.$$

Now consider

$$\limsup_{k\to\infty} A_k = \bigcap_{j=1}^{\infty} \bigcup_{k=j}^{\infty} A_k.$$

For every j,

$$\mu\left(\bigcup_{k=j}^{\infty} A_k\right) \leq \sum_{k=j}^{\infty} \mu(A_k) \leq \sum_{k=j}^{\infty} \frac{1}{2^k} \to 0.$$

Hence

$$\mu\left(\limsup_{k\to\infty} A_k\right) = 0.$$

If $x \notin \limsup_{k\to\infty} A_k$, then x belongs to only finitely many of the sets A_k. Therefore, for all sufficiently large k,

$$|f_{n_k}(x) - f(x)| \leq \frac{1}{k},$$

which implies $f_{n_k}(x) \to f(x)$. Thus $f_{n_k} \to f$ almost everywhere. $\square$

Theorem 7 (Egoroff's Theorem)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots$ be real-valued measurable functions on X. If $\mu(X) < \infty$ and*

$$f_n \to f$$

almost everywhere, then for every $\varepsilon > 0$ there exists a set $B \in \mathcal{A}$ such that

$$\mu(B^c) < \varepsilon$$

and $f_n \to f$ uniformly on B.

Proof. Fix $\varepsilon > 0$. For each n, define

$$g_n := \sup_{j \geq n} |f_j - f|.$$

Since $f_n \to f$ almost everywhere, we have

$$g_n \to 0$$

almost everywhere. Since $\mu(X) < \infty$, Proposition 3.1.2 implies that $g_n \to 0$ in measure.

Hence, for each k, we can choose n_k such that

$$\mu\left(\left\{x \in X : g_{n_k}(x) > \frac{1}{k}\right\}\right) < \frac{\varepsilon}{2^k}.$$

Define

$$B_k := \left\{x \in X : g_{n_k}(x) \leq \frac{1}{k}\right\}, \qquad B := \bigcap_{k=1}^{\infty} B_k.$$

Then

$$B^c = \bigcup_{k=1}^{\infty} B_k^c,$$

so by subadditivity,

$$\mu(B^c) \leq \sum_{k=1}^{\infty} \mu(B_k^c) < \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.$$

It remains to show uniform convergence on B. Let $\delta > 0$ and choose k such that $1/k < \delta$. Since $B \subset B_k$, for every $x \in B$ we have

$$g_{n_k}(x) \leq \frac{1}{k} < \delta.$$

Thus, for every $n \geq n_k$ and every $x \in B$,

$$|f_n(x) - f(x)| \leq g_{n_k}(x) < \delta.$$

Therefore $f_n \to f$ uniformly on B. □

Proposition 14. *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. If $f_n \to f$ in mean, i.e.*

$$\int |f_n - f| \, d\mu \to 0,$$

then $f_n \to f$ in measure.

Proof. For every $\varepsilon > 0$, Proposition 2.3.10 inequality gives

$$\mu\big(\{x \in X : |f_n(x) - f(x)| > \varepsilon\}\big) \le \frac{1}{\varepsilon}\int |f_n - f|\,d\mu.$$

Since the right-hand side tends to 0, we get

$$\mu\big(\{x \in X : |f_n(x) - f(x)| > \varepsilon\}\big) \to 0.$$

Thus $f_n \to f$ in measure. $\square$

Proposition 15 (Propositions 3.1.5)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. Suppose that $f_n \to f$ almost everywhere or in measure. If there exists an integrable function $g : X \to [0, \infty]$ such that*

$$|f_n| \le g \quad \textit{and} \quad |f| \le g$$

almost everywhere, then $f_n \to f$ in mean.

Proof. First suppose that $f_n \to f$ almost everywhere. Then

$$|f_n - f| \to 0$$

almost everywhere, and

$$|f_n - f| \le |f_n| + |f| \le 2g$$

almost everywhere. Since $2g$ is integrable, the Dominated Convergence Theorem implies

$$\int |f_n - f|\,d\mu \to 0.$$

Hence $f_n \to f$ in mean.

Now suppose instead that $f_n \to f$ in measure. By Proposition 3.1.3, every subsequence of $\{f_n\}$ has a further subsequence that converges to f almost everywhere. By the first part, every such further subsequence converges to f in mean.

If f_n did not converge to f in mean, then there would exist $\varepsilon > 0$ and a subsequence $\{f_{n_k}\}$ such that

$$\int |f_{n_k} - f|\,d\mu \ge \varepsilon$$

for all k. But this subsequence has a further subsequence converging to f almost everywhere, and hence in mean by the first part, a contradiction. Therefore $f_n \to f$ in mean. $\square$

Proposition 16 (Propositions 3.1.6)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $f, f_1, f_2, \dots \in L^1(X, \mathcal{A}, \mu)$. Suppose that $f_n \to f$ almost everywhere or in measure. If there exists an integrable function $g : X \to [0, \infty]$ such that*

$$|f_n| \le g \quad \textit{for all } n, \qquad |f| \le g$$

almost everywhere, then $f_n \to f$ in mean, i.e. $\int |f_n - f|\,d\mu \to 0$.

Proof. First suppose that $f_n \to f$ almost everywhere. Then

$$|f_n - f| \to 0$$

almost everywhere. Moreover,

$$|f_n - f| \leq |f_n| + |f| \leq 2g$$

almost everywhere. Since $2g$ is integrable, the Dominated Convergence Theorem implies

$$\int |f_n - f| \, d\mu \to 0.$$

Thus $f_n \to f$ in mean. Now suppose instead that $f_n \to f$ in measure. By Proposition 3.1.3, every subsequence of $\{f_n\}$ has a further subsequence converging to f almost everywhere. By the first part, every such further subsequence converges to f in mean.

If f_n did not converge to f in mean, then there would exist $\varepsilon > 0$ and a subsequence $\{f_{n_k}\}$ such that

$$\int |f_{n_k} - f| \, d\mu \geq \varepsilon$$

for every k. But this subsequence has a further subsequence converging to f almost everywhere, hence in mean by the first part. This is a contradiction. Therefore $f_n \to f$ in mean. □

Definition 14. *Let V be a real vector space. A function $\|\cdot\| : V \to [0, \infty)$ is called a* norm *if, for all $x, y \in V$ and $\alpha \in \mathbb{R}$,*

(a) $\|x\| = 0$ *if and only if* $x = 0$*,*

(b) $\|\alpha x\| = |\alpha| \|x\|$*,*

(c) $\|x + y\| \leq \|x\| + \|y\|$*.*

A vector space with a norm is called a normed vector space.

Definition 15. *Let V be a real vector space. A function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ is called an* inner product *if, for all $x, y, z \in V$ and $\alpha, \beta \in \mathbb{R}$,*

(a) $\langle x, x \rangle \geq 0$*, with equality if and only if* $x = 0$*,*

(b) $\langle x, y \rangle = \langle y, x \rangle$*,*

(c) $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$*.*

It induces the norm

$$\|x\| := \sqrt{\langle x, x \rangle}.$$

Definition 16. *Let $(X, \mathcal{A}, \mu)$ be a measure space and let $1 \leq p < \infty$. Define*

$$\mathcal{L}^p(X, \mathcal{A}, \mu) := \left\{ f : X \to \mathbb{R} \text{ measurable} : \int |f|^p \, d\mu < \infty \right\}.$$

For $f \in \mathcal{L}^p$, define

$$\|f\|_p := \left(\int |f|^p \, d\mu \right)^{1/p}.$$

Definition 17. *Let*

$$\mathcal{N}^p(X, \mathcal{A}, \mu) := \{f \in \mathcal{L}^p(X, \mathcal{A}, \mu) : \|f\|_p = 0\}.$$

For $1 \leq p < \infty$, this is precisely the set of functions in $\mathcal{L}^p$ that vanish almost everywhere.

Definition 18. *The space $L^p(X, \mathcal{A}, \mu)$ is defined as the quotient space*

$$L^p(X, \mathcal{A}, \mu) := \mathcal{L}^p(X, \mathcal{A}, \mu)/\mathcal{N}^p(X, \mathcal{A}, \mu).$$

Equivalently, elements of L^p are equivalence classes of functions in $\mathcal{L}^p$, where

$$f \sim g \quad \Longleftrightarrow \quad f - g \in \mathcal{N}^p \quad \Longleftrightarrow \quad f = g \text{ almost everywhere.}$$

Remark 5. *If $\langle f \rangle$ denotes the equivalence class of f, then*

$$\langle f \rangle + \langle g \rangle := \langle f + g \rangle, \qquad \alpha\langle f \rangle := \langle \alpha f \rangle.$$

Moreover, if $f \sim g$, then $\|f\|_p = \|g\|_p$, so the formula

$$\|\langle f \rangle\|_p := \|f\|_p$$

is well-defined. This gives a norm on L^p.

Definition 19. *For $p = 2$, the space $L^2(X, \mathcal{A}, \mu)$ has the inner product*

$$\langle f, g \rangle := \int fg \, d\mu.$$

The induced norm is

$$\|f\|_2 = \left(\int |f|^2 \, d\mu \right)^{1/2}.$$

Lemma 2 (Lemma 3.3.1 Young's Inequality)**.** *Let $1 < p < \infty$, and let q be defined by*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

If $x, y \geq 0$, then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

Proof. If $x = 0$ or $y = 0$, the result is clear. Assume therefore that $x, y > 0$. It is enough to prove that, for all $u, v > 0$,

$$u^{1/p} v^{1/q} \leq \frac{u}{p} + \frac{v}{q},$$

since we may then take $u = x^p$ and $v = y^q$. Let $t = u/v$. Then the desired inequality becomes

$$t^{1/p} \leq \frac{t}{p} + \frac{1}{q}.$$

Consider

$$\varphi(t) = \frac{t}{p} + \frac{1}{q} - t^{1/p}, \qquad t > 0.$$

Then

$$\varphi'(t) = \frac{1}{p} - \frac{1}{p} t^{1/p-1},$$

so $\varphi'(t) = 0$ precisely when $t = 1$. Moreover, φ has its minimum at $t = 1$, and

$$\varphi(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0.$$

Thus $\varphi(t) \geq 0$ for all $t > 0$, proving the result. $\square$

Proposition 17 (Proposition 3.3.2 Hölder's Inequality)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let p and q satisfy*

$$1 \leq p \leq +\infty, \qquad 1 \leq q \leq +\infty, \qquad \frac{1}{p} + \frac{1}{q} = 1.$$

If $f \in \mathcal{L}^p(X, \mathcal{A}, \mu)$ and $g \in \mathcal{L}^q(X, \mathcal{A}, \mu)$, then $fg \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ and

$$\int |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

Proof. First suppose that $p = 1$ and $q = +\infty$. Since

$$|g(x)| \leq \|g\|_\infty$$

holds almost everywhere, we have

$$|f(x)g(x)| \leq \|g\|_\infty |f(x)|$$

almost everywhere. Hence $fg \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ and

$$\int |fg| \, d\mu \leq \int \|g\|_\infty |f| \, d\mu = \|g\|_\infty \|f\|_1.$$

The case $p = +\infty$ and $q = 1$ is similar.

Now suppose that $1 < p < +\infty$, and hence $1 < q < +\infty$. By Young's Inequality, Lemma 3.3.1,

$$|f(x)g(x)| \leq \frac{1}{p}|f(x)|^p + \frac{1}{q}|g(x)|^q$$

for each $x \in X$. Hence, if $\|f\|_p = 1$ and $\|g\|_q = 1$, then $fg \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ and

$$\int |fg| \, d\mu \leq \frac{1}{p} \int |f|^p \, d\mu + \frac{1}{q} \int |g|^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1.$$

If neither $\|f\|_p$ nor $\|g\|_q$ is zero, apply the previous inequality to

$$\frac{f}{\|f\|_p} \qquad \text{and} \qquad \frac{g}{\|g\|_q}.$$

Then

$$\frac{1}{\|f\|_p \|g\|_q} \int |fg| \, d\mu \leq 1,$$

and hence

$$\int |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

If $\|f\|_p = 0$ or $\|g\|_q = 0$, then $fg = 0$ almost everywhere, and the inequality is clear. $\square$

Proposition 18 (Proposition 3.3.3 Minkowski's Inequality). *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let p satisfy $1 \le p \le +\infty$. If f and g belong to $\mathcal{L}^p(X, \mathcal{A}, \mu)$, then $f+g$ belongs to $\mathcal{L}^p(X, \mathcal{A}, \mu)$ and*

$$\|f+g\|_p \le \|f\|_p + \|g\|_p.$$

Proof. First suppose that $p = +\infty$. Define

$$N_1 = \{x \in X : |f(x)| > \|f\|_\infty\}, \qquad N_2 = \{x \in X : |g(x)| > \|g\|_\infty\}.$$

Then N_1 and N_2 are locally μ-null, and

$$|f(x) + g(x)| \le |f(x)| + |g(x)| \le \|f\|_\infty + \|g\|_\infty$$

holds for each x outside the locally μ-null set $N_1 \cup N_2$. Thus $f + g \in \mathcal{L}^\infty(X, \mathcal{A}, \mu)$ and

$$\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty.$$

Next suppose that $p = 1$. Then the inequality

$$|f(x) + g(x)| \le |f(x)| + |g(x)|$$

holds at each $x \in X$, and so $f + g \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ and

$$\|f+g\|_1 = \int |f+g|\, d\mu \le \int |f|\, d\mu + \int |g|\, d\mu = \|f\|_1 + \|g\|_1.$$

Now consider the case where $1 < p < +\infty$. Since

$$|f+g|^p \le (2\max\{|f|, |g|\})^p \le 2^p(|f|^p + |g|^p),$$

we have $f + g \in \mathcal{L}^p(X, \mathcal{A}, \mu)$. Define q by

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Since $p + q = pq$, it follows that

$$\left(|f+g|^{p-1}\right)^q = |f+g|^p$$

and hence that $|f+g|^{p-1} \in \mathcal{L}^q(X, \mathcal{A}, \mu)$. Thus, using the triangle inequality

$$\begin{gathered} |f+g| \le |f| + |g| \implies \\ |f+g||f+g|^{p-1} \le (|f|+|g|)|f+g|^{p-1} \implies \\ |f+g|^p \le (|f|+|g|)|f+g|^{p-1} \end{gathered}$$

and applying Proposition 3.3.2, Hölder's Inequality, to the functions f and $|f+g|^{p-1}$, and to the functions g and $|f+g|^{p-1}$, we get

$$\int |f+g|^p\, d\mu \le \int (|f|+|g|)|f+g|^{p-1}\, d\mu$$

$$= \int |f||f+g|^{p-1}\, d\mu + \int |g||f+g|^{p-1}\, d\mu$$

$$
\begin{aligned}
&\leq \|f\|_p \left\| |f+g|^{p-1} \right\|_q + \|g\|_p \left\| |f+g|^{p-1} \right\|_q \\
&= (\|f\|_p + \|g\|_p) \left(\int |f+g|^p \, d\mu \right)^{1/q}.
\end{aligned}
$$

If

$$
\int |f+g|^p \, d\mu \neq 0,
$$

we can divide by

$$
\left(\int |f+g|^p \, d\mu \right)^{1/q},
$$

obtaining

$$
\|f+g\|_p \leq \|f\|_p + \|g\|_p.
$$

Since this inequality is clear if $\int |f+g|^p \, d\mu = 0$, the proof is complete. $\square$

Theorem 8 (Completeness of L^p)**.** *Let $(X, \mathcal{A}, \mu)$ be a measure space, and let $1 \leq p \leq \infty$. Then $L^p(X, \mathcal{A}, \mu)$ is complete under the norm $\|\cdot\|_p$. Equivalently, L^p is a Banach space.*

Proof. It is enough to show that every absolutely convergent series in L^p is convergent. So let $\{f_k\}$ be a sequence in $\mathcal{L}^p$ such that

$$
\sum_{k=1}^{\infty} \|f_k\|_p < \infty.
$$

First consider the case $p = \infty$. For each k, let

$$
N_k := \{x \in X : |f_k(x)| > \|f_k\|_\infty\}.
$$

Then each N_k is locally null. Outside $\bigcup_k N_k$ we have

$$
|f_k(x)| \leq \|f_k\|_\infty
$$

for every k, and hence the series $\sum_k f_k(x)$ converges absolutely. Define

$$
f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & x \notin \bigcup_k N_k, \\ 0, & x \in \bigcup_k N_k. \end{cases}
$$

Then f is measurable and essentially bounded. Moreover, outside the locally null set $\bigcup_k N_k$,

$$
\left| f(x) - \sum_{k=1}^{n} f_k(x) \right| \leq \sum_{k=n+1}^{\infty} \|f_k\|_\infty.
$$

Therefore

$$
\left\| f - \sum_{k=1}^{n} f_k \right\|_\infty \leq \sum_{k=n+1}^{\infty} \|f_k\|_\infty \to 0.
$$

Thus the series converges in L^∞.

Now suppose $1 \leq p < \infty$. Define

$$
g(x) := \left(\sum_{k=1}^{\infty} |f_k(x)| \right)^p,
$$

with the convention that $(+\infty)^p = +\infty$.

By Proposition 3.3.3, Minkowski's Inequality, for each n,

$$\left(\int \left(\sum_{k=1}^{n} |f_k|\right)^p d\mu\right)^{1/p} = \left\|\sum_{k=1}^{n} |f_k|\right\|_p \leq \sum_{k=1}^{n} \|f_k\|_p \leq \sum_{k=1}^{\infty} \|f_k\|_p.$$

By the Monotone Convergence Theorem,

$$\int g\, d\mu = \lim_{n\to\infty} \int \left(\sum_{k=1}^{n} |f_k|\right)^p d\mu \leq \left(\sum_{k=1}^{\infty} \|f_k\|_p\right)^p < \infty.$$

Hence g is integrable. By Corollary 2.3.14, an integrable function is finite almost everywhere, so

$$\sum_{k=1}^{\infty} |f_k(x)| < \infty$$

for almost every x. Therefore the series $\sum_k f_k(x)$ converges absolutely almost everywhere.

Define

$$f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & \text{if } \sum_{k=1}^{\infty} |f_k(x)| < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Then f is measurable. Since

$$|f(x)|^p \leq g(x)$$

almost everywhere, we get $f \in \mathcal{L}^p$.

Finally, for almost every x,

$$\left|\sum_{k=1}^{n} f_k(x) - f(x)\right|^p \to 0,$$

and

$$\left|\sum_{k=1}^{n} f_k(x) - f(x)\right|^p \leq g(x).$$

Since g is integrable, the Dominated Convergence Theorem gives

$$\left\|\sum_{k=1}^{n} f_k - f\right\|_p^p = \int \left|\sum_{k=1}^{n} f_k - f\right|^p d\mu \to 0.$$

Thus

$$\left\|\sum_{k=1}^{n} f_k - f\right\|_p \to 0.$$

Hence every absolutely convergent series in L^p converges, and therefore L^p is complete.

$\square$

Chapter 5

Definition 20. *Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces. A subset of $X \times Y$ is called a* rectangle with measurable sides *if it has the form*

$$A \times B, \qquad A \in \mathcal{A}, \quad B \in \mathcal{B}.$$

The σ-algebra on $X \times Y$ generated by all rectangles with measurable sides is called the product σ-algebra *and is denoted by*

$$\mathcal{A} \times \mathcal{B}.$$

Definition 21. *Let $E \subset X \times Y$. For each $x \in X$ and $y \in Y$, the* sections *of E are defined by*

$$E_x := \{y \in Y : (x, y) \in E\}, \qquad E^y := \{x \in X : (x, y) \in E\}.$$

Definition 22. *If $f : X \times Y \to [-\infty, \infty]$ is a function, then the* sections *of f are the functions $f_x : Y \to [-\infty, \infty]$ and $f^y : X \to [-\infty, \infty]$ defined by*

$$f_x(y) := f(x, y), \qquad f^y(x) := f(x, y).$$

Lemma 3. *Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces.*

(a) If $E \in \mathcal{A} \times \mathcal{B}$, then

$$E_x \in \mathcal{B} \quad \text{and} \quad E^y \in \mathcal{A}$$

for all $x \in X$ and $y \in Y$.

(b) If $f : X \times Y \to [-\infty, \infty]$ is $\mathcal{A} \times \mathcal{B}$-measurable, then

$$f_x \text{ is } \mathcal{B}\text{-measurable} \quad \text{and} \quad f^y \text{ is } \mathcal{A}\text{-measurable}.$$

Idea. For (a), fix $x \in X$ and let $\mathcal{F}$ be the collection of all sets $E \subset X \times Y$ such that $E_x \in \mathcal{B}$. This collection is a σ-algebra and contains every measurable rectangle $A \times B$, since

$$(A \times B)_x = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A. \end{cases}$$

Hence $\mathcal{A} \times \mathcal{B} \subset \mathcal{F}$. The proof for E^y is similar.

Part (b) follows from (a), since level sets of f_x and f^y are sections of level sets of f. □

Remark 6. *The product σ-algebra is constructed so that measurable rectangles are measurable, and then closed under countable set operations. Sections are important because they allow us to study a set or function on $X \times Y$ by fixing one variable at a time. This is the basic mechanism behind product measures, Tonelli's theorem, and Fubini's theorem.*

Theorem 9 (Product measure)**.** *Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces. Then there exists a unique measure $\mu \times \nu$ on $\mathcal{A} \times \mathcal{B}$ such that*

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$$

for all $A \in \mathcal{A}$ and $B \in \mathcal{B}$. Moreover, for every $E \in \mathcal{A} \times \mathcal{B}$,

$$(\mu \times \nu)(E) = \int_X \nu(E_x)\, \mu(dx) = \int_Y \mu(E^y)\, \nu(dy).$$

Idea. For $E \in \mathcal{A} \times \mathcal{B}$, Proposition 5.1.3 says that the functions

$$x \mapsto \nu(E_x), \qquad y \mapsto \mu(E^y)$$

are measurable. Hence the formulas

$$(\mu \times \nu)_1(E) := \int_X \nu(E_x)\,\mu(dx), \qquad (\mu \times \nu)_2(E) := \int_Y \mu(E^y)\,\nu(dy)$$

define candidates for the product measure.

For disjoint measurable sets $E_n \subset X \times Y$, the sections are disjoint and

$$\left(\bigcup_n E_n\right)_x = \bigcup_n (E_n)_x.$$

Thus countable additivity of ν gives

$$\nu\left(\left(\bigcup_n E_n\right)_x\right) = \sum_n \nu((E_n)_x).$$

Using the additivity properties of the integral, this implies that $(\mu \times \nu)_1$ is a measure. Similarly, $(\mu \times \nu)_2$ is a measure.

For rectangles,

$$(A \times B)_x = \begin{cases} B, & x \in A, \\ \emptyset, & x \notin A, \end{cases}$$

so

$$(\mu \times \nu)_1(A \times B) = \int_X \nu((A \times B)_x)\,\mu(dx) = \int_X \nu(B)\mathbf{1}_A(x)\,\mu(dx) = \mu(A)\nu(B).$$

The same holds for $(\mu \times \nu)_2$. Uniqueness follows from the Dynkin-class uniqueness theorem, since the measures agree on all measurable rectangles. $\square$

Remark 7. *The main problem in constructing product measure is the following. On measurable rectangles it is clear what the measure should be:*

$$(\mu \times \nu)(A \times B) = \mu(A)\nu(B).$$

However, the product σ-algebra $\mathcal{A} \times \mathcal{B}$ contains many sets that are not rectangles, but are obtained from rectangles by countable set operations. Thus it is not enough to define the measure only on rectangles.

The purpose of the product measure construction is to extend the rectangle rule to all sets in $\mathcal{A} \times \mathcal{B}$ in a consistent and countably additive way. The section formula

$$(\mu \times \nu)(E) = \int_X \nu(E_x)\,\mu(dx)$$

explains the idea: measure each vertical slice E_x in Y, and then integrate these slice-measures over X.

The technical difficulty is proving that this gives a well-defined measure on the whole product σ-algebra and that it is the unique measure agreeing with the rectangle rule. This uniqueness step is where Dynkin classes enter.

Theorem 10 (Tonelli's Theorem)**.** *Let $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$ be σ-finite measure spaces, and let*

$$f : X \times Y \to [0, \infty]$$

be $\mathcal{A} \times \mathcal{B}$-measurable. Then:

(a) the functions

$$x \mapsto \int_Y f_x \, d\nu \qquad \text{and} \qquad y \mapsto \int_X f^y \, d\mu$$

are measurable;

(b) one has

$$\int_{X \times Y} f \, d(\mu \times \nu) = \int_X \left(\int_Y f_x \, d\nu \right) d\mu = \int_Y \left(\int_X f^y \, d\mu \right) d\nu.$$

Proof. First suppose that $f = \mathbf{1}_E$ for some $E \in \mathcal{A} \times \mathcal{B}$. Then

$$f_x = \mathbf{1}_{E_x}, \qquad f^y = \mathbf{1}_{E^y}.$$

Hence

$$\int_Y f_x \, d\nu = \nu(E_x), \qquad \int_X f^y \, d\mu = \mu(E^y).$$

By Proposition 5.1.3, the functions

$$x \mapsto \nu(E_x), \qquad y \mapsto \mu(E^y)$$

are measurable. By the construction of product measure, Theorem 5.1.4,

$$(\mu \times \nu)(E) = \int_X \nu(E_x) \, d\mu = \int_Y \mu(E^y) \, d\nu.$$

Thus the theorem holds for characteristic functions.

By linearity and homogeneity of the integral, the result also holds for every nonnegative simple measurable function

$$s = \sum_{i=1}^{n} a_i \mathbf{1}_{E_i}.$$

Indeed, applying the characteristic-function case to each E_i and summing gives

$$\int_{X \times Y} s \, d(\mu \times \nu) = \int_X \left(\int_Y s_x \, d\nu \right) d\mu = \int_Y \left(\int_X s^y \, d\mu \right) d\nu.$$

Now let f be an arbitrary nonnegative $\mathcal{A} \times \mathcal{B}$-measurable function. By Proposition 2.1.8, there exists an increasing sequence of nonnegative simple measurable functions $\{s_n\}$ such that

$$s_n \uparrow f$$

pointwise. For each fixed $x \in X$ and $y \in Y$, we then have

$$(s_n)_x \uparrow f_x, \qquad (s_n)^y \uparrow f^y.$$

By the Monotone Convergence Theorem, Theorem 2.4.1,

$$\int_Y (s_n)_x \, d\nu \uparrow \int_Y f_x \, d\nu, \qquad \int_X (s_n)^y \, d\mu \uparrow \int_X f^y \, d\mu.$$

Since each function

$$x \mapsto \int_Y (s_n)_x \, d\nu$$

is measurable, the pointwise limit

$$x \mapsto \int_Y f_x \, d\nu$$

is measurable. Similarly,

$$y \mapsto \int_X f^y \, d\mu$$

is measurable.

Applying the simple-function case to s_n gives

$$\int_{X\times Y} s_n \, d(\mu \times \nu) = \int_X \left(\int_Y (s_n)_x \, d\nu \right) d\mu.$$

Letting $n \to \infty$ and applying the Monotone Convergence Theorem on both sides yields

$$\int_{X\times Y} f \, d(\mu \times \nu) = \int_X \left(\int_Y f_x \, d\nu \right) d\mu.$$

The same argument gives

$$\int_{X\times Y} f \, d(\mu \times \nu) = \int_Y \left(\int_X f^y \, d\mu \right) d\nu.$$

Hence the theorem follows. $\square$